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Food Trade in Egypt

Analytics Graduation Project

 Final Project Report

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Table of Contents

#	Section
1	Executive Summary
2	Introduction
3	Project Objectives
4	Key Questions
5	Project Planning & Management
6	Literature Review
7	Stakeholder Analysis
8	Requirements Gathering
9	System Analysis & Design
10	Analytical Questions & Insights
11	Dashboards (Analytical Story Sequence)
12	Results & Recommendations
13	User Manual
14	References
15	Appendices

1. Executive Summary

Food Trade in Egypt Analytics Project

DEPI R4 Program | May 2026

Project Overview

This project focuses on transforming raw international food trade data into actionable, decision-ready insights that support food security policies and strategic trade decision-making in Egypt. Through systematic data cleaning, analytical modeling, and interactive visualization, the project bridges the gap between complex trade datasets and clear, policy-relevant intelligence.

Dataset Size:

- 46,691 rows before data cleaning → 45,375 rows after data cleaning
- 378 food products, 178 partner countries, spanning 6 years (2018–2023)

Key Objectives Achieved

#	Objective
1	Assess Egypt's food trade balance and export coverage ratio
2	Analyze trade partner concentration risk using the Herfindahl–Hirschman Index (HHI)
3	Identify top export products and strategically significant import items
4	Measure dependency on imported grains, with a focus on wheat and maize
5	Identify high-growth food products with strong investment potential
6	Compare export market diversification with import source concentration
7	Forecast the export coverage ratio through 2027 under multiple scenarios
8	Develop nine interactive dashboards across three analytical tools

Key Findings

- 1- Persistent food trade deficit of -\$53.5B (2018–2023).
- 2- Export coverage ratio improved from 36% to 49.6%.
- 3- High grain dependency (94% imported).
- 4- Import concentration risk (39.9% from top 3 countries).
- 5- Strong export diversification (178 partners, HHI = 0.033).

Main Risks Identified

Risk	Level	Impact
Import concentration (3 countries = 39.9%)	● High	Geopolitical supply chain vulnerability
Meat production gap dependency (99% imported)	● High	Food insecurity if global prices spike
Grain dependency (94% imported)	● High	Structural production deficit

Opportunities Identified

Opportunity	Potential
Fruits & Vegetables exports	Competitive advantage (77%, 96% export share)
Chickens, Yoghurt, Eggs	Early signals for VC & agri-investment
Agro-processing	Value-added export potential

Forecast

Scenario	Description	Coverage Ratio by 2027
Optimistic	Agro-processing + import diversification	62-65%
Base Case	Continuation of current trend	58-62%
Pessimistic	Geopolitical or climate shock	50-52%

Strategic Recommendations

#	Recommendation	Priority	Responsible Body
1	Diversify grain import sources from 3 to 6+ countries (Kazakhstan, Romania, France)	● Urgent	Ministry of Supply
2	Invest in agro-processing for fruits & vegetables	● High	Ministry of Agriculture
3	Develop local meat and poultry production	● High	Ministry of Agriculture
4	Establish strategic grain reserves (6+ months)	● Medium	Ministry of Supply
5	Create monthly monitoring system for key indicators	● Continuous	CAPMAS

Tools & Deliverables

Tool	Number of Dashboards
Excel	3
Tableau	4
Power BI	2
Total	9 Dashboards

Conclusion

Egypt's food trade shows improving resilience, as the export coverage ratio increased from 36% in 2018 to 49.6% in 2023. However, persistent vulnerabilities remain due to high import concentration, significant grain dependency, and structural gaps in meat production. Strengthening food security therefore requires targeted actions focusing on import diversification, agro-processing expansion, and enhanced local production to reduce long-term trade deficits.

2. Introduction

2.1 Project Idea & Data Source

This project analyzes Egypt's food trade during the period 2018–2023 using real, non-synthetic data obtained from the Food and Agriculture Organization (FAO) database. The dataset covers all food product categories and trading partners related to Egypt's import and export activities, providing a comprehensive basis for analysis.

Dataset Size:

Metric	Value
Rows before cleaning	46,691
Rows after cleaning	45,375
Food products	378
Partner countries	178
Years	6 (2018-2023)

2.2 Why This Topic?

#	Reason
1	Based on real (non-synthetic) data
2	Directly focused on Egypt
3	Food security is a national priority
4	Provides actionable insights for decision-makers

2.3 Project Scope

Dimension	Scope
Geographic	Egypt (Reporter country: Egypt)
Temporal	2018–2023 (six years)
Thematic	All food products (imports & exports)

3. Project Objectives

#	Objective
1	Assess Egypt's food trade balance and export coverage ratio
2	Analyze key trade partners and concentration risks
3	Identify the most impactful products in exports and imports
4	Measure dependency on imported grains (wheat and maize)
5	Identify promising export opportunities based on high-growth products
6	Compare export market diversification with import source concentration
7	Forecast the export coverage ratio until 2027
8	Build an interactive dashboard summarizing the complete food trade picture

4. Key Questions

4.1 Strategic Questions (7 Questions)

#	Question
Q1	How is Egypt's food trade balance evolving, and is the coverage ratio improving?
Q2	Who are the most important trade partners, and is there a concentration risk?
Q3	Which products drive exports, and which place the highest burden on imports?
Q4	How dependent is Egypt on imported grains (Wheat and Maize)?
Q5	Which products show dramatic growth spikes as early signals for investment?
Q6	How diversified are export markets compared to import sources?
Q7	What is the forecast for the export coverage ratio by 2027?

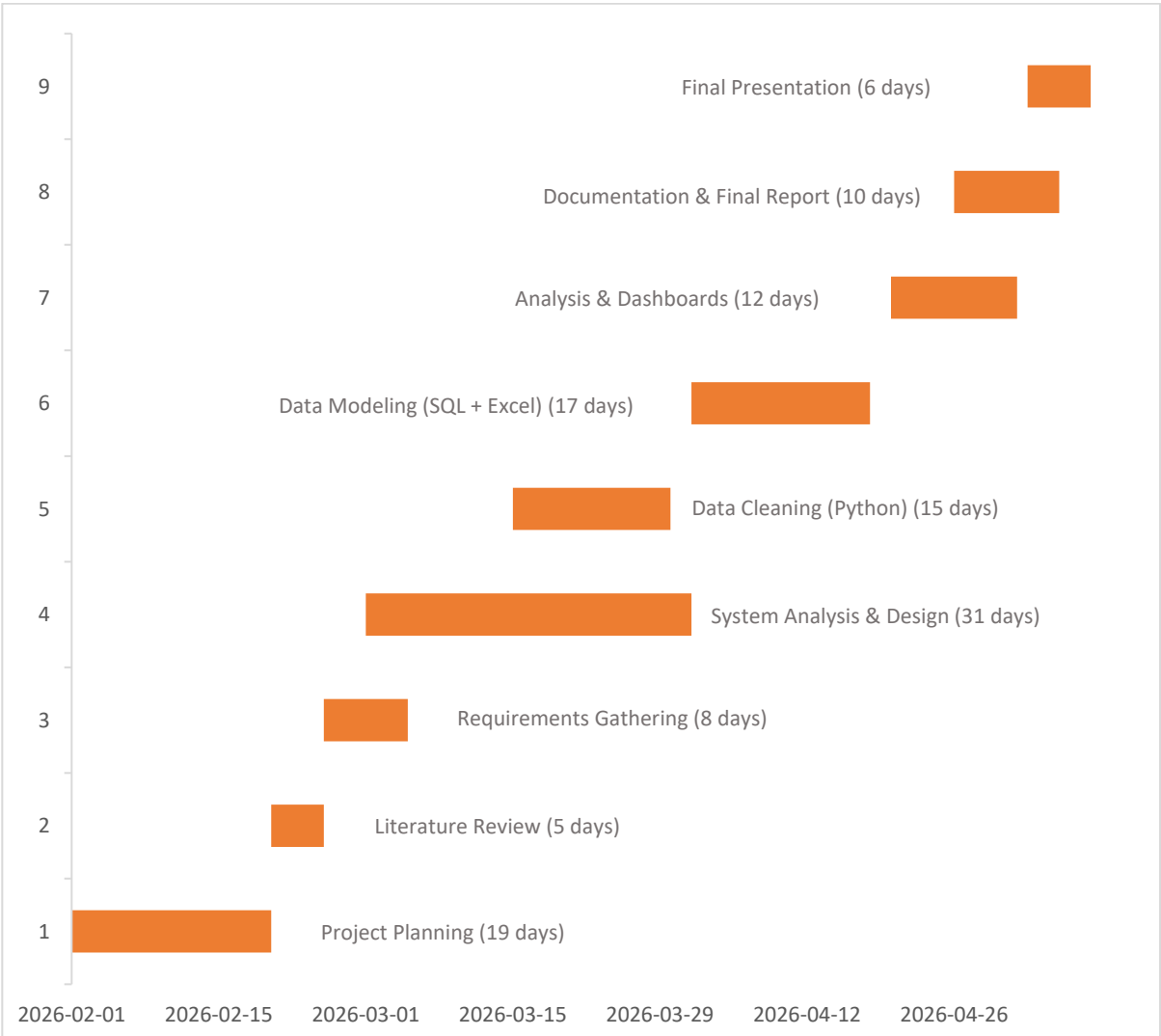
4.2 Analytical Questions per Dashboard (9 Dashboards)

#	Dashboard	Question
1	Trade Balance & Coverage	Is there structural improvement in Egypt's trade balance and coverage ratio?
2	Categories: Risk vs Opportunity	Which categories represent export opportunities versus import risks?
3	Top Products & Partners	Which products and partners are the largest and fastest-growing?
4	Partner Risk & Concentration	Are import partners more concentrated and riskier than export partners?
5	Growth & Opportunities Matrix	Which products combine high growth with high stability?
6	Why Scenario Forecasting?	Why is forecasting needed (shocks, deficits, and deviations)?
7	Forecasting & Scenarios	When can the coverage ratio reach 60%, and under which scenario?
8	Categories Trends Over Time	How does each category's share evolve over time?
9	Interactive Executive Dashboard	What is the complete picture, and where are risks concentrated?

5. Project Planning & Management

5.1 Timeline (Gantt Chart)

Phase	Start	End	Duration
Project Planning	1/2/2026	20/2/2026	20 days
Literature Review	20/2/2026	25/2/2026	5 days
Requirements Gathering	25/2/2026	5/3/2026	10 days
System Analysis & Design	1/3/2026	1/4/2026	30 days
Data Cleaning (Python)	15/3/2026	30/3/2026	15 days
Data Modeling (SQL + Excel)	1/4/2026	18/4/2026	18 days
Analysis & Dashboards	20/4/2026	2/5/2026	12 days
Documentation & Final Report	26/4/2026	6/5/2026	10 days
Final Presentation	3/5/2026	9/5/2026	6 days



5.2 Task Assignment

Phase	Tasks	Responsible Members
Data Cleaning (Python)	Defining data cleaning strategy and rules	All members
	Python script execution and final implementation	
Data Modeling (SQL)	Database design, star schema, and ETL processes	All members
Data Modeling (Excel)	Initial calculations and pivot tables	All members
Questions & Insights	Developing analytical questions and extracting insights	All members
Dashboard Development		
	Dashboards 1, 2 (Excel)	All members
	Dashboard 3 (Excel)	
	Dashboard 4 (Tableau)	
	Dashboards 5, 6, 7 (Tableau)	
	Dashboards 8, 9 (Power BI)	
Documentation	Report writing and formatting	All members
Team Leadership	Project management and coordination	Amina Adel Soliman

5.3 Risk Assessment & Mitigation Plan

#	Risk	Probability	Impact	Mitigation
1	Large dataset size (46,691 rows)	High	Medium	Python was used instead of Excel to ensure efficient processing and scalability
2	Missing quantity values (3,809 rows)	High	High	Applied a price-ratio-based imputation algorithm to estimate missing quantities
3	Inconsistent measurement units	Medium	Medium	Standardized all units using Python during the data cleaning phase
4	Tool incompatibility	Low	Medium	Ensured version alignment across tools and conducted regular backups
5	Time constraints	Medium	High	Developed a realistic project schedule with built-in time buffers

5.4 Key Performance Indicators (KPIs)

#	KPI	Target	Achieved
1	Number of dashboards developed	6-9	✓ 9
2	Imputation success rate	>60%	✓ 65.5%
3	Number of generated insights	>30	✓ 40+
4	Number of tools used	3	✓ 3
5	Complete project documentation	100%	✓

6. Literature Review

Source	Topic	Relevance
FAO (2023)	Food trade statistics	Primary data source
World Bank (2024)	Egypt Economic Monitor	Data validation
CAPMAS (2024)	Egyptian trade statistics	Macroeconomic interpretation

Research Gap Filled by This Project:

No previous study combined:

This project addresses a clear research and practice gap by integrating several components that have not been jointly explored in previous studies. Specifically, no prior work combined large-scale data cleaning using Python, star schema modeling implemented through SQL Server, and multi-tool analytical workflows utilizing Excel, Power BI, and Tableau within a single coherent framework. In addition, this project uniquely delivers nine integrated dashboards designed to collectively answer seven strategic, policy-relevant questions, thereby bridging the gap between raw trade data, advanced analytics, and decision-ready insights.

7. Stakeholder Analysis

Stakeholder	Primary Interest	Needs from Project
Ministry of Supply	Food security and import policy design	Recommendations for supplier diversification and risk reduction
Ministry of Agriculture	Export development and market expansion	Identification of high-potential export products and target markets
Investors Association	Investment opportunities in agri-food sectors	High-growth, stable, and scalable products
Academic researchers	Methodological rigor and reproducibility	A transparent and replicable analytical framework
DEPI Jury	Project quality and learning outcomes	Compliance with graduation project requirements

8. Requirements Gathering

8.1 Functional Requirements

#	Requirement
FR-1	The system can import and clean FAO trade data
FR-2	The system can calculate the Herfindahl–Hirschman Index (HHI) to assess market concentration
FR-3	The system can compute the export coverage ratio
FR-4	The system can perform time-series trend analysis
FR-5	The system can forecast the export coverage ratio up to 2027
FR-6	The system can export analytical results to interactive dashboards

8.2 Non-Functional Requirements

#	Requirement	Specification
NFR-1	Query performance	System response time less than 5 seconds for standard queries
NFR-2	Data accuracy	Error rate maintained below 1% after data cleaning and validation
NFR-3	Scalability	Ability to seamlessly incorporate additional years of data
NFR-4	Tool compatibility	Full compatibility with Excel, Power BI, and Tableau
NFR-5	Documentation quality	Complete and well-structured code and project documentation

9. System Analysis & Design

9.1 Problem Statement

Raw FAO food trade data is unstructured and contains missing values, inconsistent measurement units, and varying data quality across countries and years, As a result, the dataset cannot be directly used for food security analysis or policy-oriented decision-making without systematic cleaning, transformation, and analytical modeling.

9.2 System Architecture & Data Flow

The system follows a structured data pipeline that transforms raw trade data into decision-ready analytical dashboards, The overall architecture is illustrated as follows:

CSV Files (FAO) → Python (Cleaning) → MS SQL (Star Schema) → Dashboards (Excel/Power BI/Tableau)

Pipeline Steps:

The analytical pipeline consists of the following stages:

- 1. Importing libraries**
Core Python libraries including Pandas and NumPy were used for data manipulation and numerical processing.
- 2. Reading FAO data**
Raw trade data was imported from FAO CSV/Excel files into the Python environment.
- 3. Exploratory data inspection**
Initial exploration was conducted to examine dataset structure, dimensions, data types, and missing values.
- 4. Data cleaning and preprocessing**
Missing values, inconsistent units, and invalid records were handled using rule-based and algorithmic cleaning methods.
- 5. Exploratory data analysis (EDA)**
Descriptive statistics and trend exploration were performed to understand trade patterns and anomalies.
- 6. KPI calculation**
Key performance indicators were computed, including the Herfindahl–Hirschman Index (HHI), export coverage ratio, and coefficient of variation (CV).
- 7. Database modeling**
Cleaned and transformed data was exported to SQL Server and organized using a star schema design to support analytical queries.
- 8. Dashboard development**
Nine interactive dashboards were developed across Excel, Power BI, and Tableau to answer seven strategic research questions.

9.3 Data Cleaning Examples (Python)

This section presents selected examples from the Python-based data cleaning process applied to the FAO dataset, highlighting key challenges and corresponding solutions

Example 1: Cleaning the Quantity Column

python: To ensure numeric consistency, the *Quantity* column was converted to numeric format, and invalid or non-positive values were treated as missing.

```
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
df.loc[df['Quantity'] <= 0, 'Quantity'] = np.nan
```

This step standardized quantity values and eliminated erroneous entries that could distort analytical results.

Example 2: Standardizing Measurement Units

python: FAO data contained quantities reported in inconsistent units, such as “1000 An”. These units were standardized to ensure comparability.

```
kilo_animal_mask = df['Quantity_Unit'] == '1000 An'
df.loc[kilo_animal_mask, 'Quantity'] = df.loc[kilo_animal_mask, 'Quantity'] * 1000
df.loc[kilo_animal_mask, 'Quantity_Unit'] = 'An'
```

This transformation ensured that all quantities were expressed using a unified base unit.

Example 3: Imputing Missing Quantities Using Price Ratios

python: Missing quantity values were estimated using a price-ratio imputation approach based on trade value and median unit prices.

Two hierarchical median levels were applied:

- Level 1: Category, Year, and Trade Flow
- Level 2: Year and Trade Flow (fallback level)

```
median_L1 = df.groupby(['Category', 'Year', 'Flow'])['Unit_Price'].transform('median')
median_L2 = df.groupby(['Year', 'Flow'])['Unit_Price'].transform('median')
final_median = median_L1.fillna(median_L2)
estimated_qty = np.where(
    (df['TradeValue_USD'] > 0) & (final_median > 0),
    (df['TradeValue_USD'] / final_median).round(2),
    np.nan)
df['Quantity'] = df['Quantity'].fillna(pd.Series(estimated_qty, index=df.index))
```

This approach maximized data retention while maintaining economic consistency.

Results:

IMPUTATION QUALITY REPORT

Total rows originally missing Quantity: 3,809

Successfully estimated (Price Ratio) : 2,493

Cannot impute (TradeValue also 0/null): 1,316

Imputation success rate : 65.5%

Stage	Row Count
Before Cleaning	46,691
After Cleaning	45,375
Dropped rows	1,316

Rows were dropped only when imputation was infeasible due to missing or zero trade values, ensuring data integrity without excessive loss.

9.4 Database Design (Star Schema)

Dimension Tables:

SQL

```
CREATE TABLE countries (  
    country_id INT IDENTITY PRIMARY KEY,  
    country_name VARCHAR(100) UNIQUE NOT NULL  
);
```

```
CREATE TABLE products (  
    product_id INT IDENTITY PRIMARY KEY,  
    item_code_fao INT NOT NULL,  
    product_name VARCHAR(255),  
    category VARCHAR(100)  
);
```

```
CREATE TABLE years (year INT PRIMARY KEY);  
CREATE TABLE trade_flows (flow_id INT IDENTITY PRIMARY KEY, flow_type VARCHAR(10));  
CREATE TABLE units (unit_id INT IDENTITY PRIMARY KEY, unit_name VARCHAR(20));  
CREATE TABLE quantity_sources (source_id INT IDENTITY PRIMARY KEY, source_type VARCHAR(20));
```

Fact Table:

```
CREATE TABLE trade_fact (  
    trade_id BIGINT IDENTITY PRIMARY KEY,  
    reporter_country_id INT NOT NULL,  
    partner_country_id INT NOT NULL,  
    product_id INT NOT NULL,  
    year INT NOT NULL,  
    flow_id INT NOT NULL,  
    unit_id INT NOT NULL,  
    source_id INT NOT NULL,  
    quantity DECIMAL(18,4),  
    trade_value_usd DECIMAL(18,2),  
    dataset_primary_key VARCHAR(255),  
    FOREIGN KEY (reporter_country_id) REFERENCES countries(country_id),  
    FOREIGN KEY (partner_country_id) REFERENCES countries(country_id),  
    FOREIGN KEY (product_id) REFERENCES products(product_id),  
    FOREIGN KEY (year) REFERENCES years(year),  
    FOREIGN KEY (flow_id) REFERENCES trade_flows(flow_id),  
    FOREIGN KEY (unit_id) REFERENCES units(unit_id),  
    FOREIGN KEY (source_id) REFERENCES quantity_sources(source_id)  
);
```

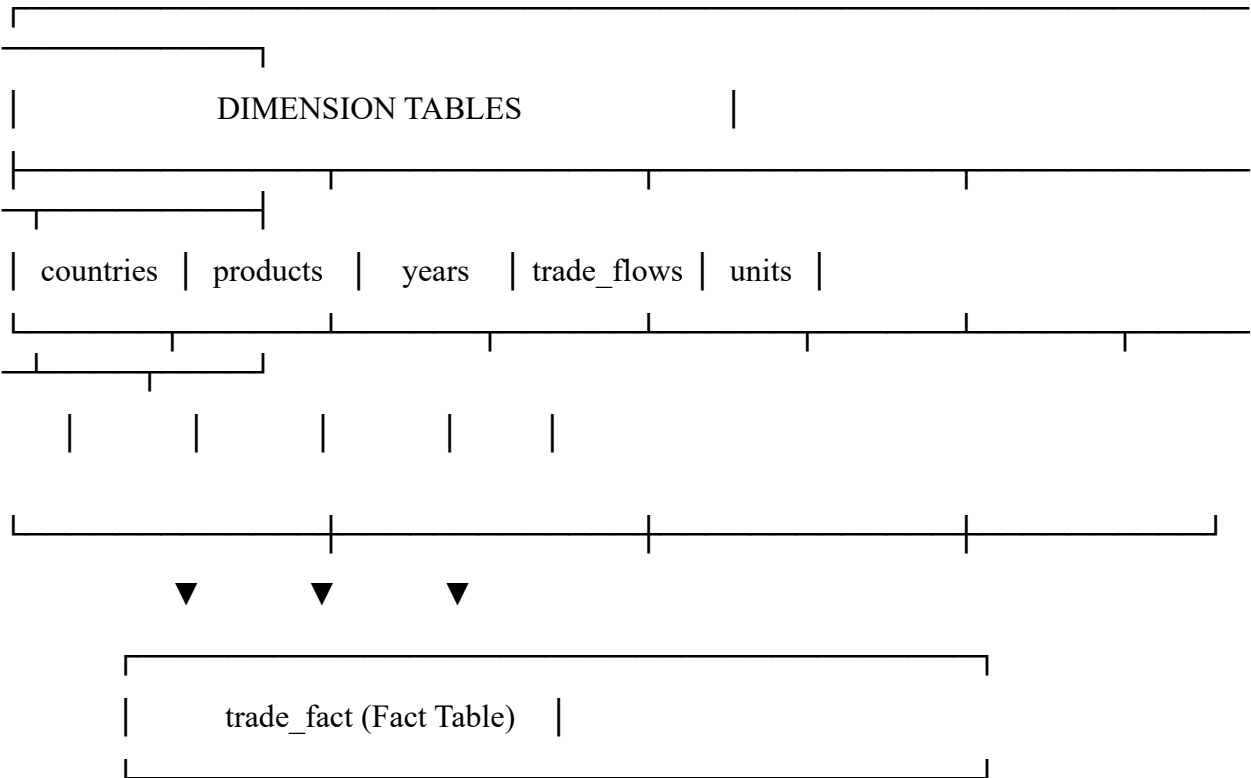
Indexes for Performance:

SQL

```
CREATE INDEX idx_fact_year ON trade_fact(year);  
CREATE INDEX idx_fact_product ON trade_fact(product_id);  
CREATE INDEX idx_fact_reporter ON trade_fact(reporter_country_id);  
CREATE INDEX idx_fact_partner ON trade_fact(partner_country_id);
```

Star Schema Diagram:

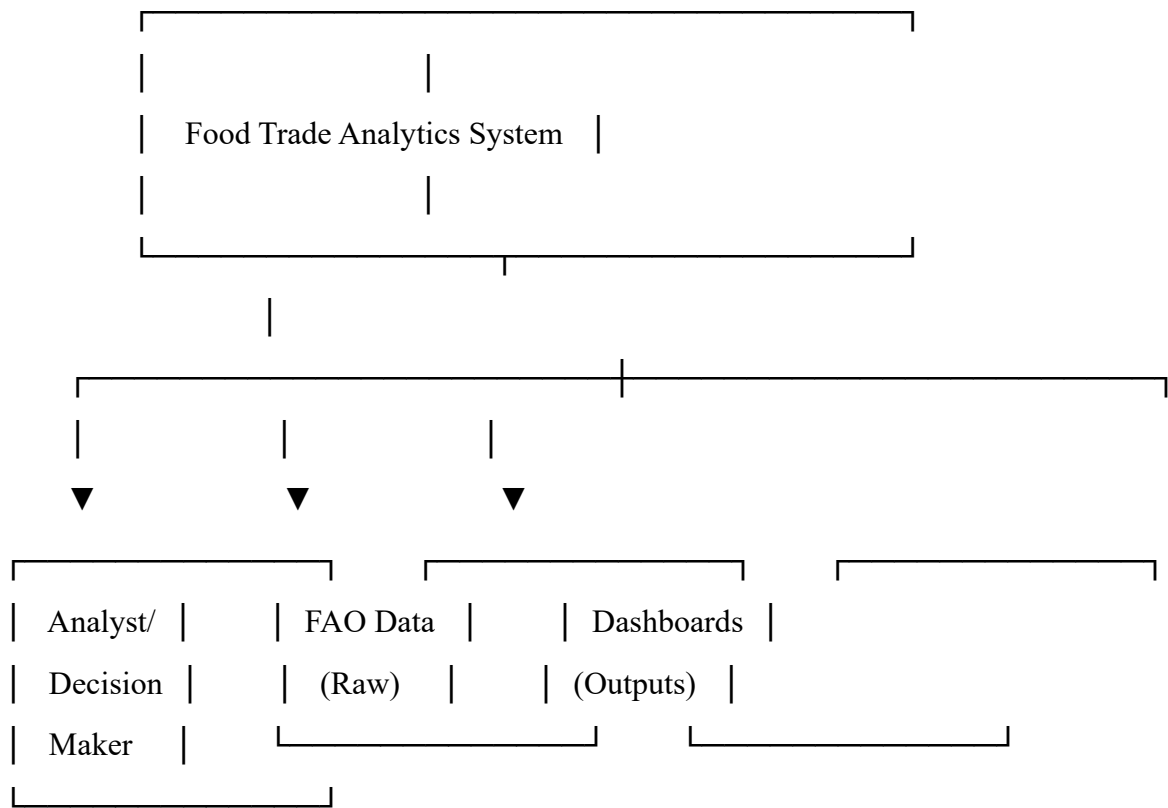
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9.5 Data Flow Diagram (DFD)

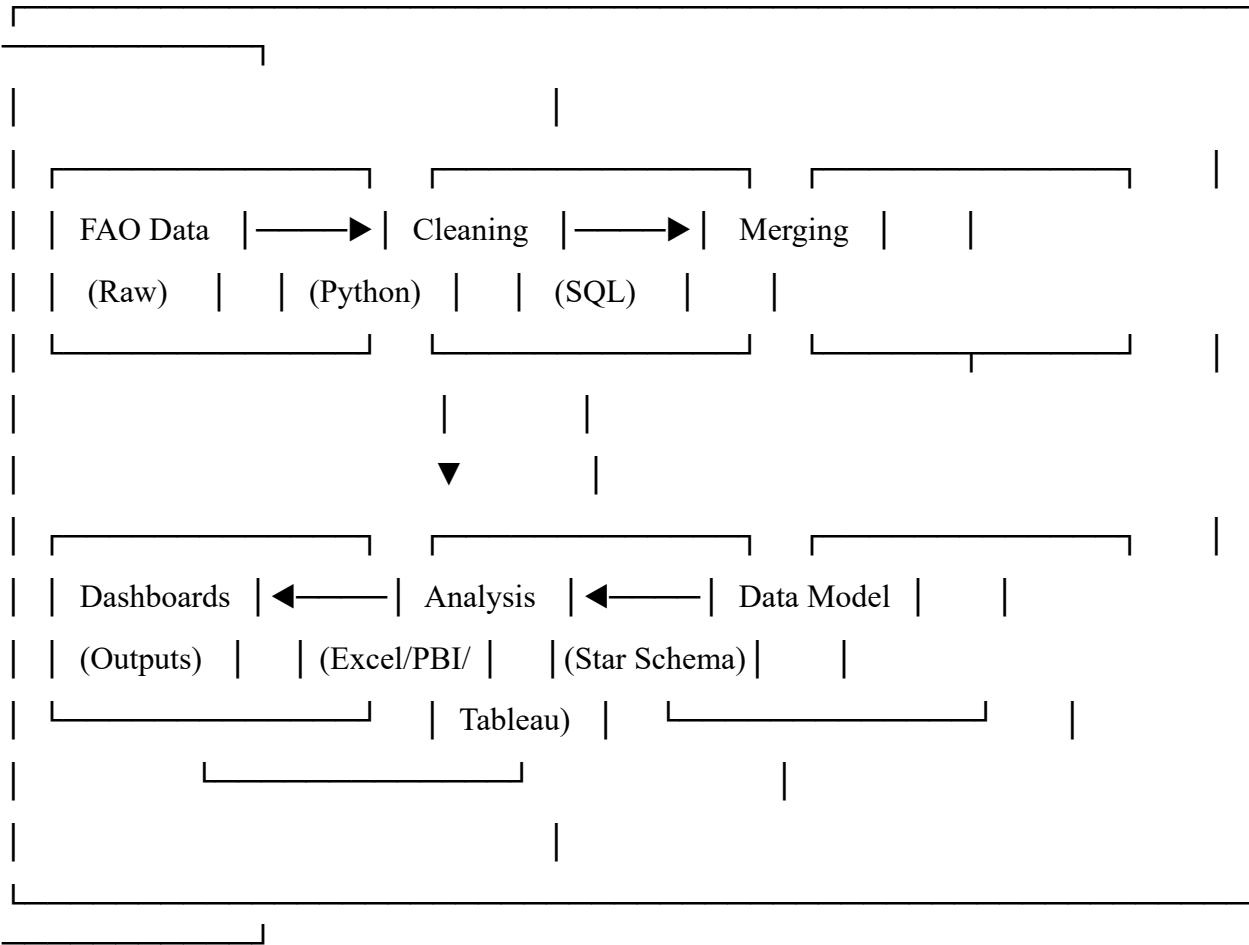
Context Level:

text



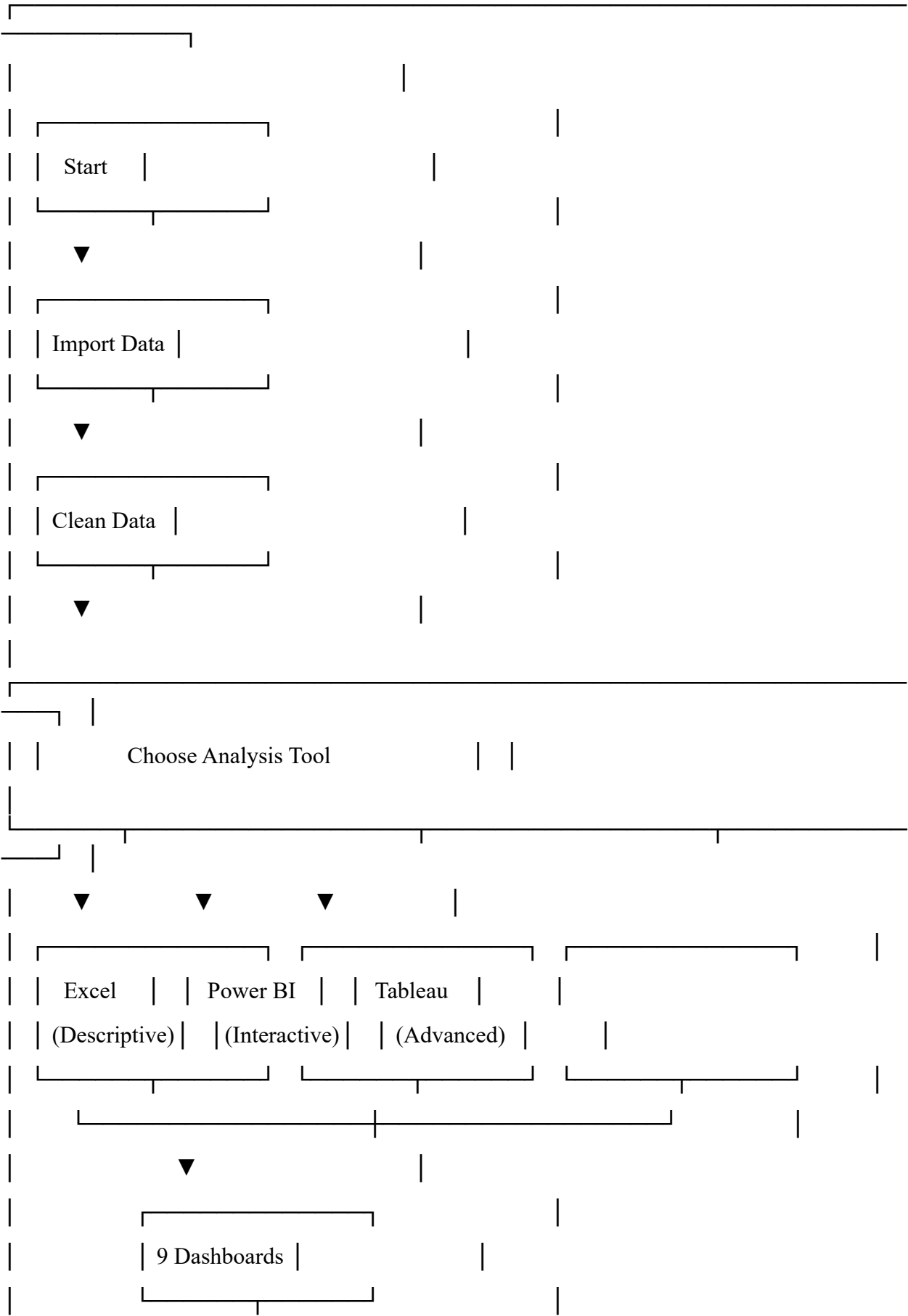
Level 1 DFD:

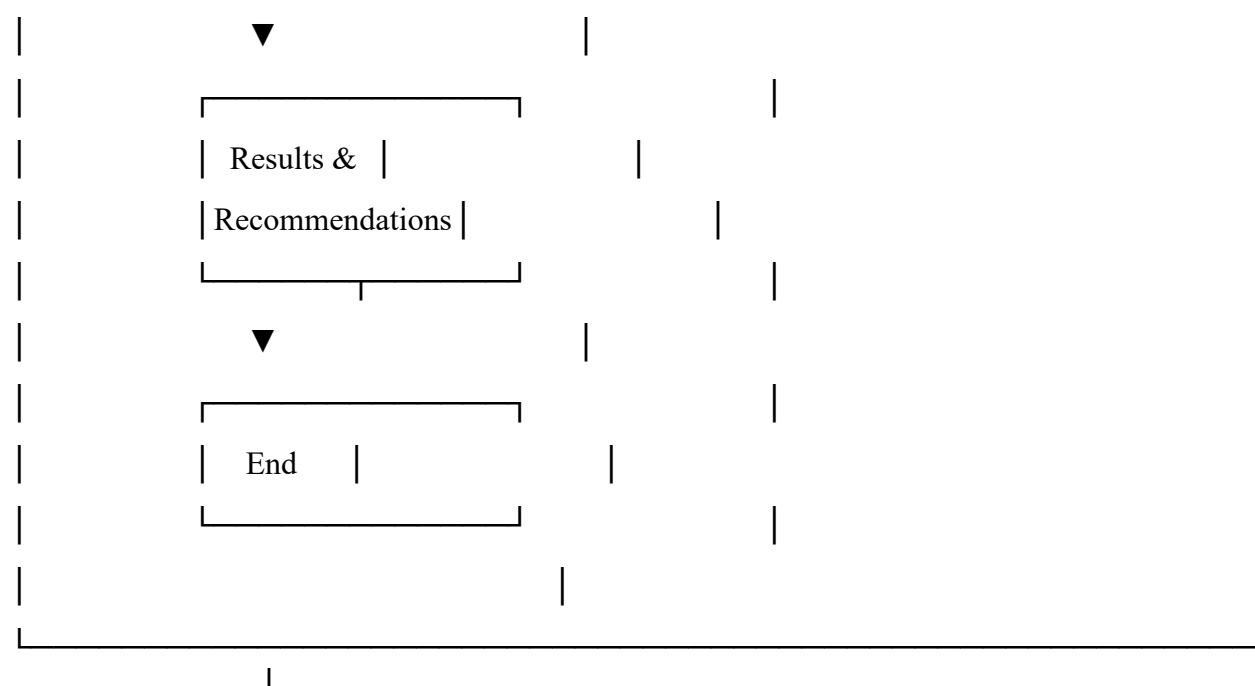
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9.6 Activity Diagram (Analysis Process)

text





9.7 Technology Stack

Layer	Technology	Version	Purpose
Data Processing	Python	3.9.7	Data cleaning, preprocessing, and imputation
Python Libraries	pandas, NumPy	latest	Data manipulation and numerical analysis
Database	SQL Server	2019	Star schema implementation and analytical storage
Modelling	Excel (Power Pivot)	2019	Initial calculations and data modeling
Visualization	Excel	2019	Development of 3 descriptive dashboards
Visualization	Power BI	Desktop	Development of 2 interactive dashboards
Visualization	Tableau	2026.1	Development of 4 advanced analytical dashboards

10. Analytical Questions & Insights

This section synthesizes the key analytical insights derived from the developed dashboards. The findings focus on trade balance dynamics, import dependency risks, export diversification patterns, and indicators of structural improvement over time.

Key Insights:

#	Insight
1	Chronic food trade deficit – Egypt exhibits a persistent food trade deficit amounting to approximately USD 53.5 billion over the period 2018–2023 , reflecting structural reliance on food imports.
2	Improving export coverage ratio – Export coverage improved from 36% in 2018 to 49.6% in 2023 , indicating a gradual but incomplete narrowing of the trade gap.
3	High grain dependency risk – Cereals are 94% import-dependent , with wheat and maize imports totaling USD 30.9 billion , representing a significant food security vulnerability.
4	Strong export market diversification – Egypt exports food products to 178 partner countries , with a low HHI of 0.033 , reflecting excellent export market diversification and limited concentration risk.

Supporting Insights:

- Meat and poultry dependency – The meat and poultry category is 99.2% import-dependent, indicating a structural domestic production gap.
- Import concentration risk – Imports are moderately concentrated, with Russia, the United States, and Brazil accounting for 39.9% of total food imports.
- Export competitive advantage – Oranges represent Egypt's largest food export, totaling approximately USD 3.98 billion, highlighting a strong competitive position in the fruits category.
- High growth spikes – Exceptional growth rates were observed in chickens (+16,800%) and yogurt (+10,154%), serving as early signals of emerging investment opportunities (noting base-effect influence).
- Forecasting outlook – Under the base-case scenario, the export coverage ratio is projected to reach 58–62% by 2027, indicating continued but gradual improvement.

11. Dashboards (Analytical Story Sequence)

The Story Flow:

Current Situation (Q1, Q3, Q4) → Risk (Q2) → Opportunity (Q5) → Future (Q7) → Complete Picture

11.1 Dashboard 1: Trade Balance & Coverage

Element	Detail
Tool	Excel
Key Question	Is there structural improvement in Egypt's trade balance and export coverage ratio?
Chart Type	Combo chart (Exports & Imports as bars, Coverage Ratio as line) with YoY growth bars
KPIs	Total exports, total imports, trade balance, export coverage ratio, year-over-year growth indicators

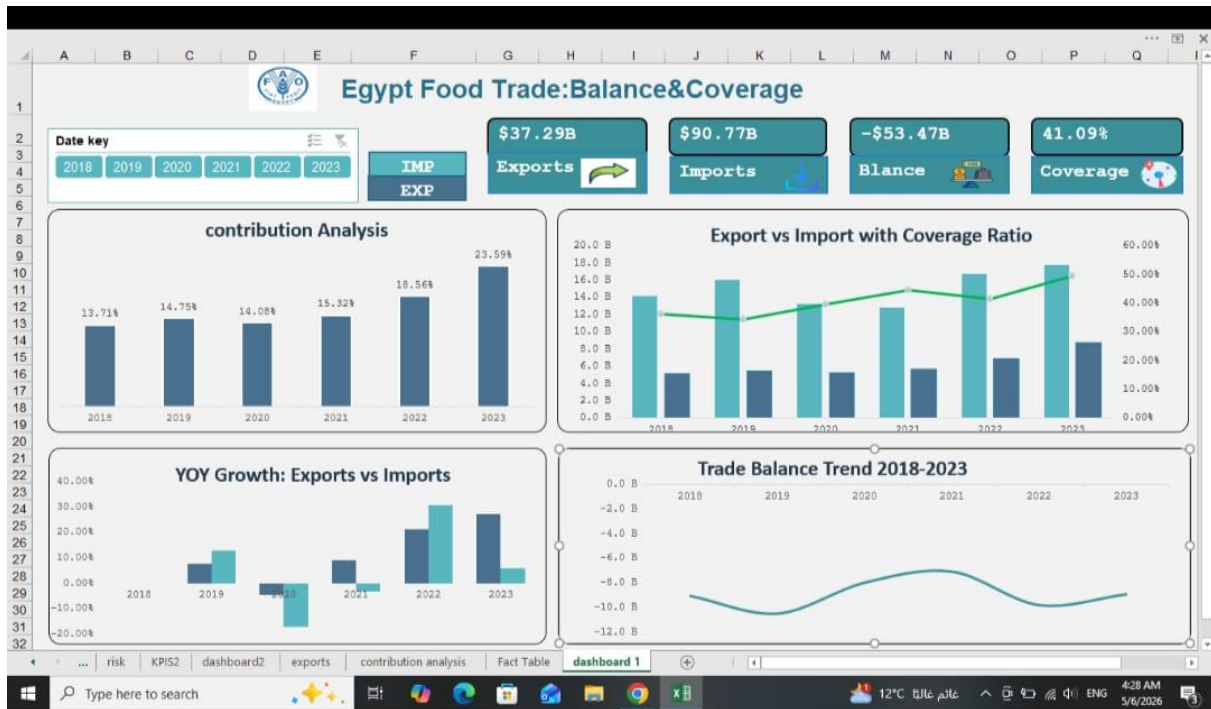
Results:

Year	Exports	Imports	Balance	Coverage
2018	\$5.11B	\$14.19B	-\$9.08B	36.0%
2019	\$5.50B	\$16.01B	-\$10.51B	34.4%
2020	\$5.25B	\$13.25B	-\$8.00B	39.6%
2021	\$5.71B	\$12.82B	-\$7.11B	44.6%
2022	\$6.92B	\$16.74B	-\$9.82B	41.3%
2023	\$8.80B	\$17.75B	-\$8.95B	49.6%

Recommendations:

1. Continue supporting food exports (+27% growth in 2023)
2. Monitor external shocks (COVID 2020, Ukraine war 2022)

Figure 1: Trade Balance & Coverage Dashboard



11.2 Dashboard 2: Categories – Risk vs Opportunity

Element	Detail
Tool	Excel
Key Question	Which categories are export opportunities vs import risks?
Chart Type	100% Stacked Bar + Grouped Bar
KPIs	Highest Export Category, Highest Import Category, Highest Dependency %

Results:

Category	Exports	Imports	Dependency	Direction
Fruits	\$9.07B	\$2.71B	23% imported	✅ Export
Vegetables	\$4.78B	\$0.19B	4% imported	✅ Export
Processed Food	\$6.40B	\$5.29B	45% imported	Balanced
Oils & Fats	\$1.61B	\$8.68B	84% imported	⚠️ Import
Cereals	\$2.05B	\$33.14B	94% imported	🔴 Risk
Meat	\$0.07B	\$9.13B	99% imported	🔴 Risk

Top 3 Import Risk Categories:

Category	Dependency
Meat & Poultry	99.2%
Cereal & Grains	94.2%
Live Animals	90.9%

Recommendations:

1. Invest in local meat production (chicken, cattle)
2. Expand processed fruit exports (agro-processing)

Figure 2: Categories – Risk vs Opportunity Dashboard



11.3 Dashboard 3: Top Products & Partners

Element	Detail
Tool	Excel
Key Question	Which are the largest and fastest-growing products & partners?
Chart Type	Horizontal Bar Chart + Variance Bars + CAGR Line
KPIs	Largest Export (Value), Largest Export (Quantity), Highest Growth %, Top Partner

Results – Top Products:

Product	Value	Rank
Oranges	\$3.98B	1 (largest export)
wheat	\$18.11B	1 (largest import)

Results – Fastest Growing Products (CAGR):

Product	CAGR (2018→2023)
Brewing or distilling dregs	261%
Chickens	179%
Persimmons	167%
Yoghurt	152%

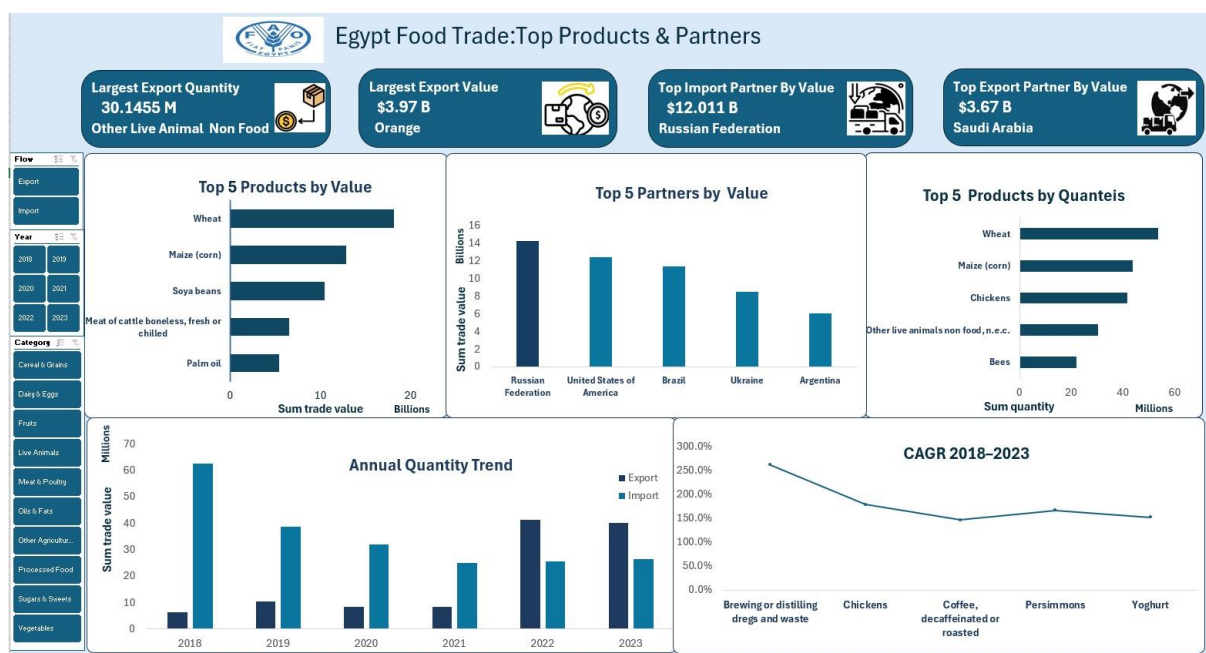
Results – Top Partners:

Partner	Role	Value
Russia	Largest import source	\$12.01B
Saudi Arabia	Largest export market	\$3.68B

Recommendations:

1. Monitor Chickens and Yoghurt for sustainable growth (base effect warning)
2. Use these "Early Signals" for investment decisions

Figure 3: Top Products & Partners Dashboard



11.4 Dashboard 4: Partner Risk & Concentration

Element	Detail
Tool	Tableau
Key Question	Are import partners riskier than export partners?
Chart Type	HHI Cards + Line Chart + Heatmap + Scatter Plot
KPIs	Import HHI, Export HHI, Top 3 Import Share, Top 3 Export Share

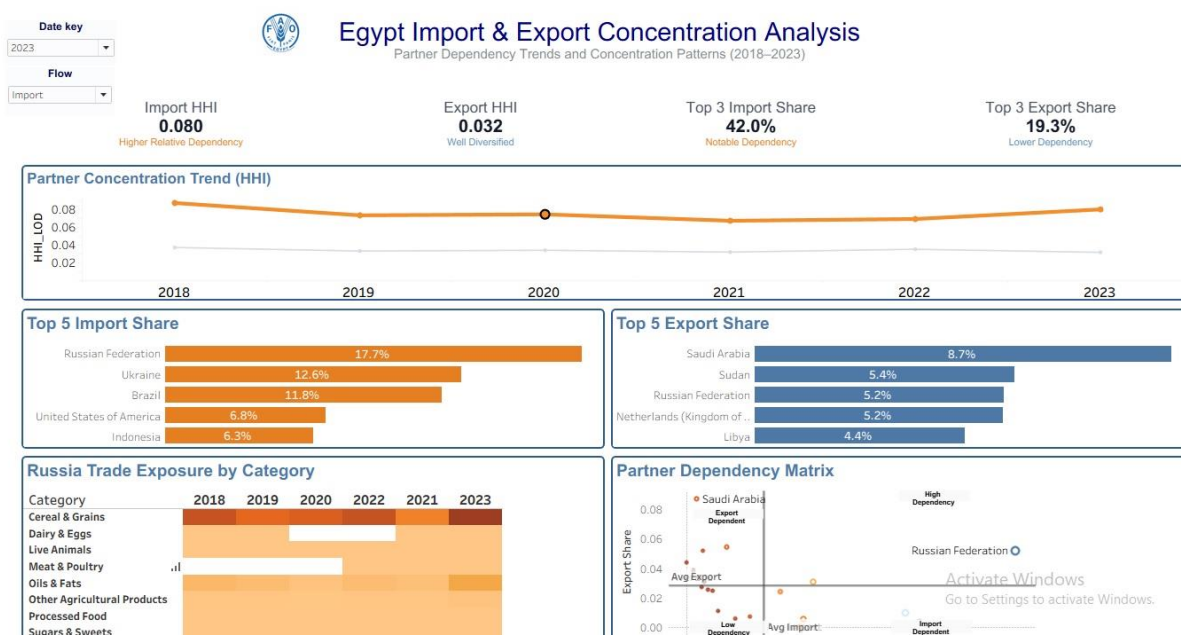
Results:

Metric	Value	Classification
Import HHI 2023	0.080	● Higher Relative Dependency
Export HHI 2023	0.032	● Well Diversified
Top 3 Import Share	42%	● Notable Dependency
Top 3 Export Share	19.3%	● Lower Dependency

Recommendations:

1. Diversify grain import sources from 3 to 6+ countries (Kazakhstan, Romania, France)
2. Leverage export market diversity (178 countries) for trade negotiations

Figure 4: Partner Risk & Concentration Dashboard



11.5 Dashboard 5: Growth & Opportunities Matrix

Element	Detail
Tool	Tableau
Key Question	Which products combine high growth and high stability?
Chart Type	Bubble Chart (4 Quadrants) + Box Plot
KPIs	Highest Growth %, Most Stable (Lowest CV), Stars Count, Average CV

Results:

Result	Value
Highest Growth Product	Chickens (+16,800%)
Most Stable Product	Oranges (CV = 0.12)
Stars Count	8 products
Average CV	0.45

The 4 Quadrants:

Quadrant	Meaning	Action
Stars 	High Growth + High Stability	Prioritize investment
Bets 	High Growth + Low Stability	Monitor carefully
Cows 	Low Growth + High Stability	Improve efficiency
Losers 	Low Growth + Low Stability	Review or discontinue

Recommendations:

- 1 Prioritize investment in "Stars" quadrant products
- 2 Monitor "Bets" products for sustainable growth
- 3 Improve efficiency for "Cows" products
- 4 Review or discontinue "Losers" products

Figure 5: Growth & Opportunities Matrix Dashboard



11.6 Dashboard 6: Why Scenario Forecasting? (Intro)

Element	Detail
Tool	Tableau
Key Question	Why is forecasting needed? (Shocks, deficits, deviations)
Chart Type	Line Chart + Bar Chart
KPIs	Trade Deficit by Year, Coverage Gap to 60%, Trade Shock Impact

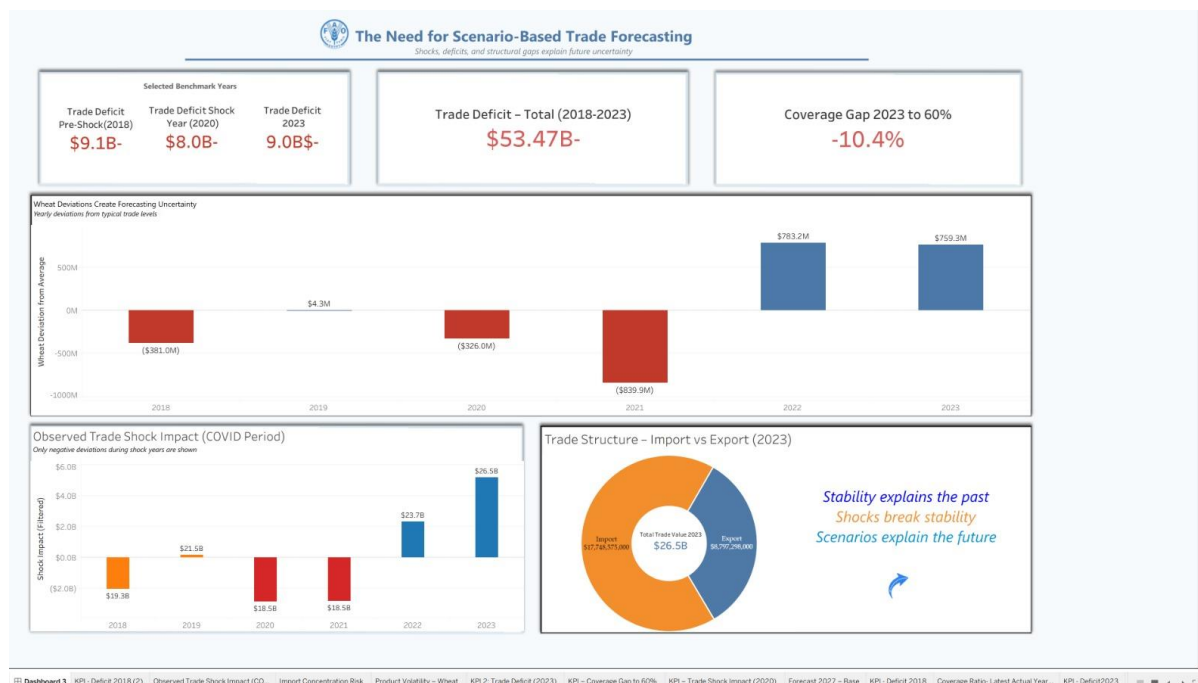
Key Messages:

- Trade deficit fluctuates with external shocks (COVID, Ukraine war)
- Coverage gap to 60% = additional exports needed
- Shocks break stability → scenarios explain the future

Recommendations:

- 1 Use scenario planning rather than single-point forecasts
- 2 Prepare contingency plans for pessimistic scenarios

Figure 6: Why Scenario Forecasting Dashboard



11.7 Dashboard 7: Forecasting & Scenarios

Element	Detail
Tool	Tableau
Key Question	When can coverage ratio reaches 60% and under which scenario?
Chart Type	Line Chart with Forecast + Multiple Scenario Lines
KPIs	Coverage Ratio 2023 (49.6%), Forecast 2027, MAPE (4.2%)

Results:

Scenario	Description	Forecast 2027
Optimistic	Agro-processing + Import diversification	62-65%
Base Case	Continuation of current trend	58-62%
Pessimistic	Geopolitical or climate shock	50-52%

Recommendations:

- 1 Diversify grain import sources to achieve optimistic scenario
- 2 Invest in agro-processing to increase export value

Figure 7: Forecasting & Scenarios Dashboard



11.8 Dashboard 8: Categories Trends Over Time

Element	Detail
Tool	Power BI
Key Question	How does each category's share change over time?
Chart Type	Line Chart (Multi-line) + 100% Stacked Bar + Drill through
KPIs	Total Exports (2023), Total Imports (2023), Fastest Growing Category

Results – Import Share Change:

Category	2018	2023	Change
Cereals	38%	34%	-4% 
Other Ag.	22%	24%	+2%
Meat	10%	10%	Stable

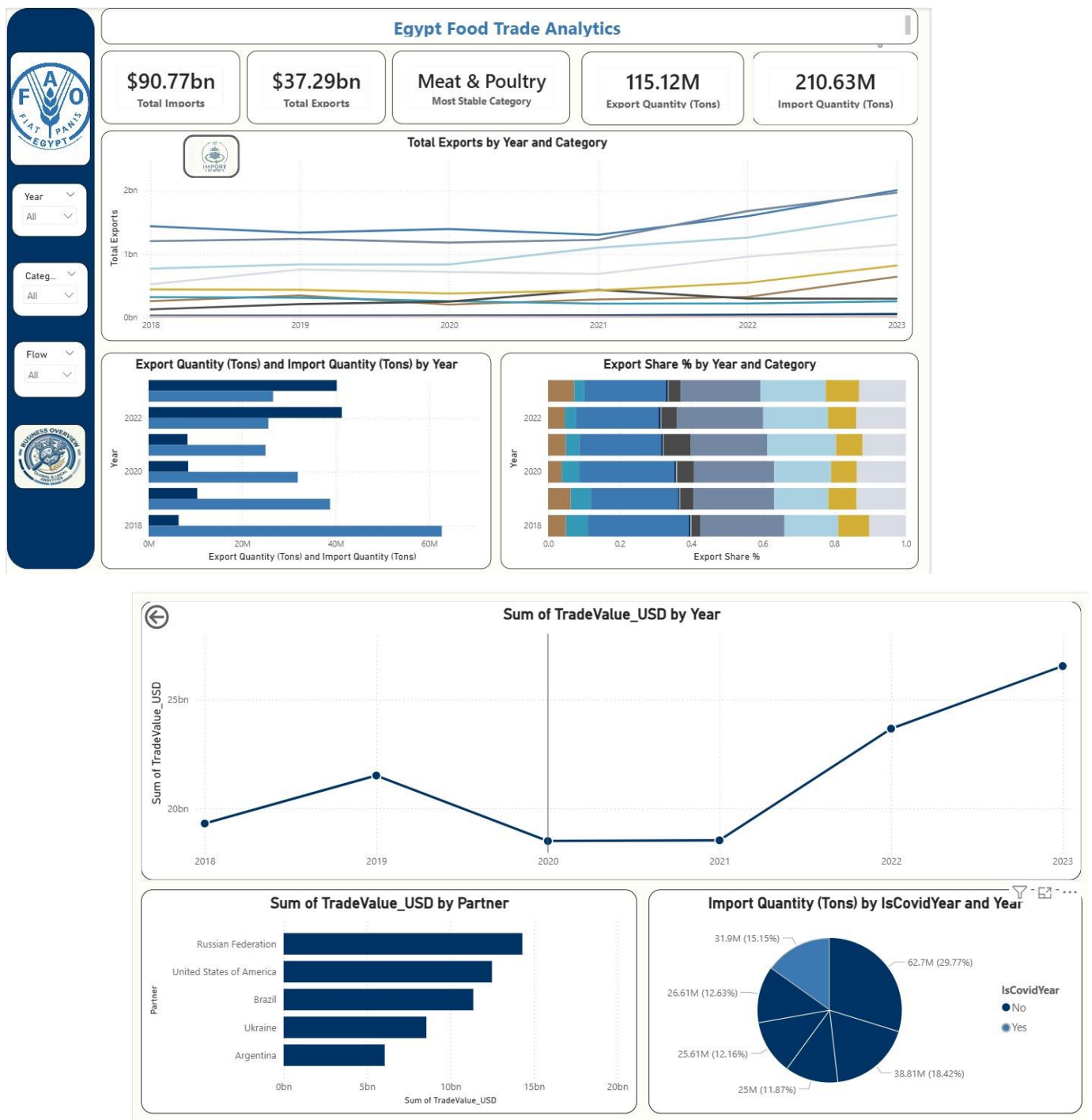
Results – Export Share Change:

Category	2018	2023	Change
Fruits	22%	24%	+2% 
Processed Food	16%	17%	+1%
Vegetables	12%	13%	+1%

Recommendations:

- 1 Accelerate export diversification beyond fruits (e.g., processed foods)
- 2 Monitor high-dependency categories quarterly

Figure 8: Categories Trends Over Time Dashboard



11.9 Dashboard 9: Interactive Executive Dashboard

Element	Detail
Tool	Power BI
Key Question	What is the complete picture and where are risks concentrated?
Chart Type	Mixed (KPI Cards + Line Chart + Bar Charts + 100% Stacked Bar)
KPIs	Total Exports, Total Imports, Trade Balance, Coverage Ratio, YoY Growth

Main KPIs:

KPI	Value	Status
Total Exports (2018-2023)	\$37.3B	
Total Imports (2018-2023)	\$90.8B	
Cumulative Trade Balance	-\$53.5B	
Coverage Ratio 2023	49.6%	
YoY Growth 2023	+27.1%	

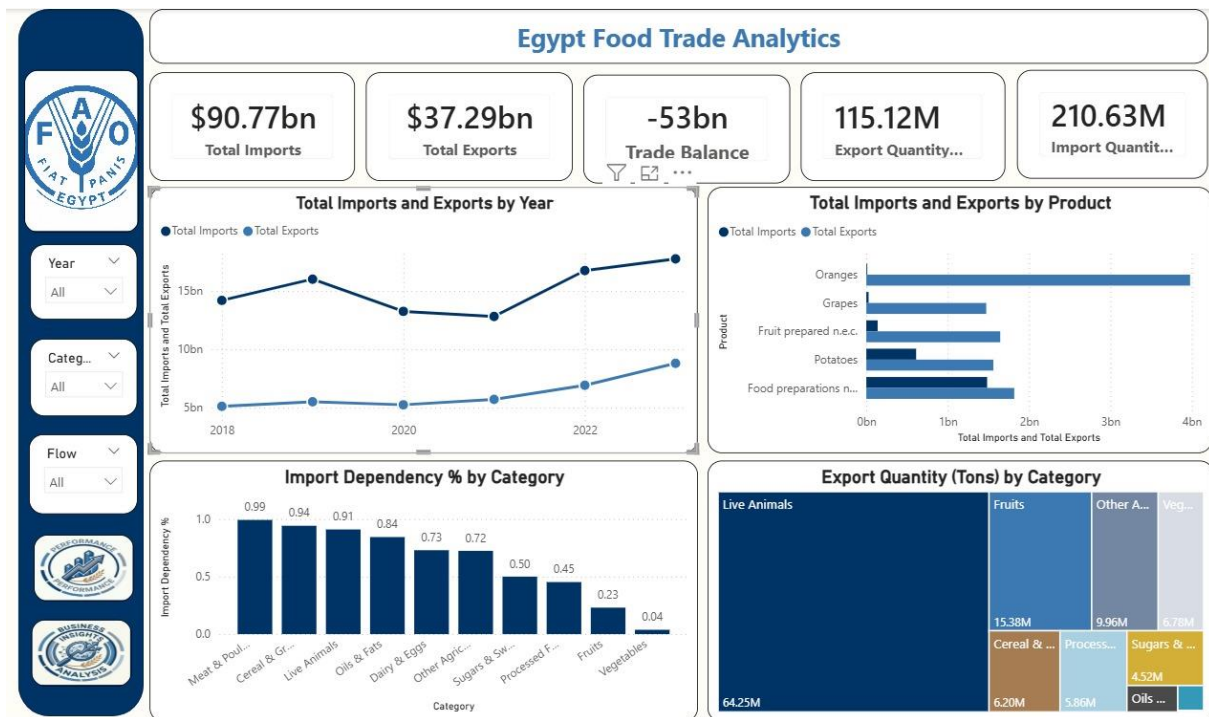
Risk Summary:

Risk	Level	Action
Import Concentration (3 countries = 39.9%)	 High	Diversify grain import sources
Grain Dependency (94% imported)	 High	Local production + strategic reserves
Meat Gap (99% imported)	 High	Livestock investment

Recommendations:

- 1 Diversify grain import sources from 3 to 6+ countries
- 2 Create monthly monitoring system for key indicators

Figure 9: Interactive Executive Dashboard



12. Results & Strategic Recommendations

12.1 Key Findings Summary

1. Chronic food trade deficit (-\$53.5B, 2018-2023)
2. Coverage ratio improved from 36% (2018) to 49.6% (2023)
3. 2023 highest export year (\$8.80B) with +27% YoY growth
4. Wheat alone = 14.1% of total trade (\$18.12B)
5. Oranges largest export (\$3.98B) – competitive advantage
6. Meat & Poultry: 99.2% imported – structural gap
7. Cereals & Grains: 94% imported – food security risk
8. Import concentration: Russia + USA + Brazil = 39.9%
9. Export diversification: 178 countries (HHI=0.033)
10. Growth spikes: Brewing or distilling dregs 261%, Chickens 179%

12.2 Strategic Recommendations

#	Recommendation	Priority	Responsible Body
1	Diversify grain import sources from 3 to 6+ countries	● Urgent	Ministry of Supply
2	Invest in agro-processing for fruits & vegetables	● High	Ministry of Agriculture
3	Develop local meat and poultry production	● High	Ministry of Agriculture
4	Establish strategic grain reserves (6+ months)	● Medium	Ministry of Supply
5	Create monthly monitoring system for key indicators	● Continuous	CAPMAS

13. User Manual (Dashboard User Guide)

13.1 How to Use Tableau Dashboards (Dashboards 4, 5, 6, 7)

Step	Action
1	Use Year Filter to select a specific year or range
2	Hover over any data point for detailed tooltips
3	Click on legend items to filter specific categories
4	Read the Insight Box at the bottom for key takeaways
5	For Dashboard 4: Use Flow Filter (Import/Export) to compare

13.2 How to Use Excel Dashboards (Dashboards 1, 2, 3)

Step	Action
1	Use Slicers (if available) to filter by Year or Category
2	Follow the color coding:  Good,  Medium,  Critical
3	Review summary tables for exact numbers
4	Check the Insight Box for recommendations

13.3 How to Use Power BI Dashboards (Dashboards 8, 9)

Step	Action
1	Use Slicers (Year, Category, Flow) to filter data
2	Click on any chart element to cross-filter other visuals
3	For Dashboard 8: Click on any Category to drill through
4	Read the Narrative Visual for automatic insight generation
5	Check Risk Flags (  ) for immediate risk assessment

14. References

1. Food and Agriculture Organization of the United Nations (FAO). (2023). *Food and agriculture data (FAOSTAT)*. <https://www.fao.org/faostat>
2. World Bank. (2024). *Egypt economic monitor*. Washington, DC.
3. Central Agency for Public Mobilization and Statistics (CAPMAS). (2024). *Egyptian foreign trade statistics*.

15. Appendices

1. Appendix A: Raw Dataset Source

FAO Food Trade Dataset (2018–2023)

Source: Food and Agriculture Organization (FAO) Database

File attached separately: raw_food_trade_data.xlsx

2. Appendix B: Python Cleaning Code

Separate file attached: data_cleaning.ipynb

3. Appendix C: SQL Scripts & Validation Queries

Separate files attached:

- SQL Schema & ETL Scripts (file: Code.txt)
- Validation Queries (file: تقرير ال سيكوال.docx)

4. Appendix D: GitHub Repository

All project source code, data files, dashboards and documentation are available at:

https://github.com/AminaAdelSaleh/Achievements-Data-Analysis/tree/main/project_cleaning_phase1